ISYS 1108 / 3474 Software Engineering Project Management

Dr Maria Spichkova

Senior Lecturer in Software Engineering

Week 1: Lectorial

What's next...

RMIT UNIVERSITY

Acknowledgement of Country

RMIT University acknowledges the people of the Woi wurrung and Boon wurrung language groups of the eastern Kulin Nation on whose unceded lands we conduct the business of the University.

RMIT University respectfully acknowledges their Ancestors and Elders, past and present.

RMIT also acknowledges the Traditional Custodians and their Ancestors of the lands and waters across Australia where we conduct our business.

"Womin Djeka"

is the Aboriginal, Kulin Nation word for...

"Welcome, what is your purpose?"

It is statement as well as a question

Our plan for this session

- 😊 Core organizational points: sessions, schedule, expectations, assessments…
- 😊 SE Project Management: Why?
- 😊 Discussion on pre-recorded materials
- 😊 Project Methodologies: overview
- 😊 Vision & Scope document vs. Project Charter
- 😊 Work Breakdown Structure (WBS): Traditional vs. Agile

+ any questions you have on related material ☺

Who?

You:

- Bachelor and Master students

Course Coordinator and Lecturer:

- Dr Maria Spichkova, Senior Lecturer in Software Engineering
  maria.spichkova@rmit.edu.au

  Feedback/ Consultation hours (might be updated later):
  - Tuesdays, 16:30 – 17:30pm, office 14.10.13

Tutors:

Anthony Clapp, Fiona Le, Anh Nguyen Thi Phuong (Sofia), Osheen Walawedura, Zhenshuo (Vincent) Zhao

Who am I?

Dr Maria Spichkova

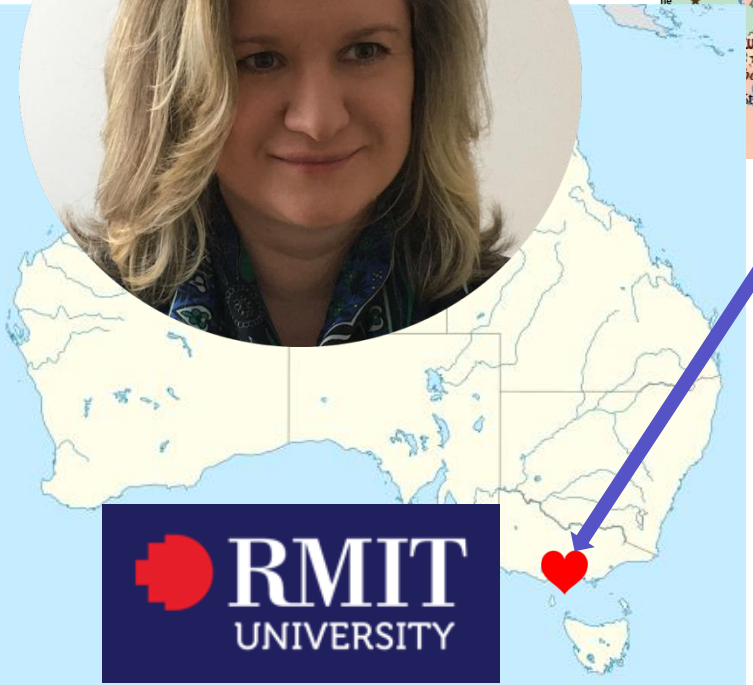
TECHNISCHE UNIVERSITÄT MÜNCHEN

TECHNISCHE UNIVERSITÄT DRESDEN

SIEMENS

DAAD Deutscher Akademischer Austausch Dienst German Academic Exchange Service

МЭИ



RMIT UNIVERSITY

Active researcher in Software Engineering: human-oriented SE, Agile, RE, modelling, testing as well as formal specification and verification of safety-critical and distributed systems.

Why are we doing this course?

This is a very common mistake to make ☺

IT graduates often think good programming skills is all they require

- What if I my interest is only into coding?
- Hmm... I don't want to be a project manager!!!!!111

IT Industry is very dynamic and is changing very fast today

You need other skills too

- Software Engineering
- Project Management
- Professional Communication and above all

This course is not about being a Project Manager, but about being an IT Professional / Software Engineer ☺

Where & When?

- Lectorials: discussion of general ideas + "big picture"+ theory + analysis + examples + inspiration

- Workshops: more analysis- and skill-based

Work on set problems + discussions + answering questions

Workshops:
Enrolment is crucial, we will allocate you to the teams within the class you are enrolled in – you cannot attend another on Campus class

Invest at least 6h in reading and learning the materials + working on the assignments

Classes and Deadlines

Mid-semester break is between Weeks 5 and 6

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Lectorials												
Workshops						Team-based assignment activities						
A1: SE Challenge				Friday: Challenge 1 10%					Friday: Challenge 2 10%		Friday: Challenge 3 10%	
A2: Team-based assignment								Due on Friday: 30%				
A3: Individual Assignment												Due on Friday: 40%

A1: SE Challenges

- ✓ Individual assessment
- ✓ 3 online challenges with a number of practical case studies with real-life scenarios: on Fridays of Weeks 4, 9, 11
- ✓ weight - 10% each,
- ✓ timed, to be completed within within 24h window – you should demonstrate that you not only can solve project management tasks but also can do it within reasonable time
- ❗ No “late submission” possible – if missed by a valid reason, the only option is a deferred challenge based on granted *special consideration (can be granted only by Spec. Cons. Office, Course Coordinator cannot provide it)*

A2: Team-based Assignment

Assignments are team-based, unless you have corresponding ELP

Simulated Scrum Project: 30%

You team will be provided with scenarios, for which you will need to create Scrum artefacts (product backlog, sprint backlog, notes, etc.)

The teams will be formed by tutors during the workshop sessions.

Typical size of a team: 4-5 students enrolled in the same workshop class.

The core work will be done in Workshop sessions 6+7, under the observation of the tutor.

A3: Individual Assignment (on-demand interview): 40%

- You will work individually on a Scrum Project scenario, to create a burn-down chart and to analyse some other artefacts. You will need to create two short videos to present your results and a report.
- Marks announced at the official release date

Extensions

- Extensions per person, not per team (even for the team-based assessment)
- Valid and documented reasons. A Course Coordinator can approve only if a valid certificate is provided:
  - Serious illness, family tragedy, etc.
  - Provide corresponding **valid certificate**
- **Extensions up to 7 calendar days:**
  - Not applicable for A1 (SE Challenges)
  - Applicable for A2: you might get an alternative assessment, to submit without your team
  - Applicable for A3: You might get an extension
  - Apply at least one working day before the deadline – apply using Canvas "Assessment extension" functionality!

Special consideration: If it is granted then

- A1: Deferred SE challenge
- A2: Alternative assessment with extension **more than 7 calendar days**
- A3: Extension **more than 7 calendar days**
- Apply within 5 working day from the deadline – follow the RMIT Special Consideration policy: https://www.rmit.edu.au/students/student-essentials/assessment-and-exams/assessment/special-consideration

Marking

Why did I lose marks…?

You don't "lose" marks, you EARN them!

- ✓ You start with a blank page and 0 marks
- ✓ **You earn marks for what you do:**
  you get rewarded for what you do <u>correctly</u>
- ✓ You don't get penalized for what you don't do
- ✓ Full marks are possible, but rare and needs to be earned

Questions

Within the teaching period (Weeks 1-12), I will try to answer your questions within 48h or less, within business hours (except weekends and public holidays ☺).

In some circumstances I might answer out of business hours, but this is an <u>exception</u>, not a rule.

Please do net expect answering your questions 24/7 ☺.

- Ask!
- Ask early!

Questions to me:

- Ask preferably during lectures and workshops – there might be more students having the same question
- Feedback/ Consultation session
- EdForum
- Emails: Write from your RMIT account, otherwise Outlook might put your message in junk ☹

Please use this link to join EdForum using your RMIT student email address

https://edstem.org/au/join/KJWMWB

If you are an exchange student or have RMIT Vietnam student email address, please send me an email and I will add you to the EdForum manually.

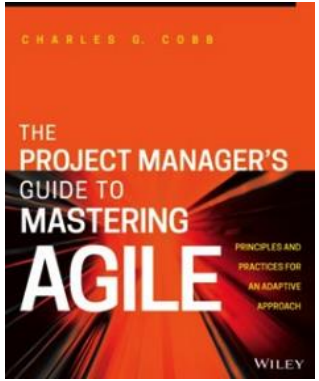
Recommended Materials

Papers on recent research and industry trends – will be shared in Canvas

+

(recommended, not compulsory)

"The Project Manager's Guide to Mastering Agile: Principles and Practices for an Adaptive Approach", Charles Cobb



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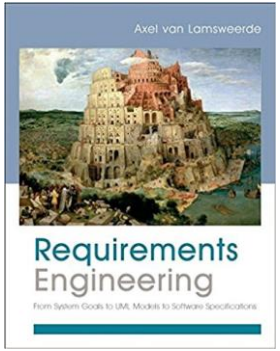
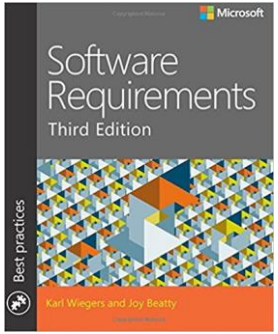
(for RE-related topics)

- "Requirements Engineering", Axel van Lamsweerde
- "Software Requirements", Karl Wiegers, Joy Beatty





Preliminary schedule “Syllabus” in Canvas

Why “preliminary”?

Because all the mentioned material is important, but we can refocus some parts so that your preferences are taken into account

Week	Lectorial Topics	Workshop topics	Assessments Due
Week 1	Motivation - why Project Management (PM) skills are important for software developers/ IT specialists? Why do we focus on Agile? What is the correlation between Requirements Engineering and Project Management? Project Charter vs. Vision & Scope Document. Project methodologies Work Breakdown Structure (WBS)		
Week 2	Agile/Scrum: User stories, Product Backlog	Project Charter vs. Vision & Scope Document. Work Breakdown Structure (WBS)	
Week 3	Agile/Scrum: Epics, User Story Cards, Acceptance criteria	User stories, Product backlog	
Week 4	Agile/Scrum: Sprint planning and meetings, Agile estimates	User story cards, Acceptance criteria	SE Challenge 1: WBS, Project planning (A1.1)
Week 5	Agile/Scrum Sprint planning, Lean and Kanban	Sprint planning notes, sprint retro notes	
	Mid-semester break between Weeks 5 and 6		
Week 6	Lean and Kanban	Trello, Completeness of a Product backlog	
Week 7	Scrum and Kanban diagrams	Jira, Kanban WIP	
Week 8	Alternatives to Agile: Hybrid methodologies, RUP. Requirements Elicitation as a background for PM	Burn-down charts	Team-based assignment on Scrum artefacts (A2)
Week 9	Alternatives to Agile: Critical path analysis	Selection of a methodology, activities and artefact for a particular project	SE challenge 2: Burn-down chart analysis, Scrum planning, Kanban (A1.2)
Week 10	Requirements as a basis for planning, Quality assurance	Critical path analysis	
Week 11	Advanced concepts in Agile, DevOps, CMMI	Traditional vs. Agile requirements. Quality assurance.	SE challenge 3: Critical path analysis, methodology selection (A1.3)
Week 12	Course summary/ revision	Advanced concepts on project progress analysis	Individual assignment on Scrum artefacts (A3)

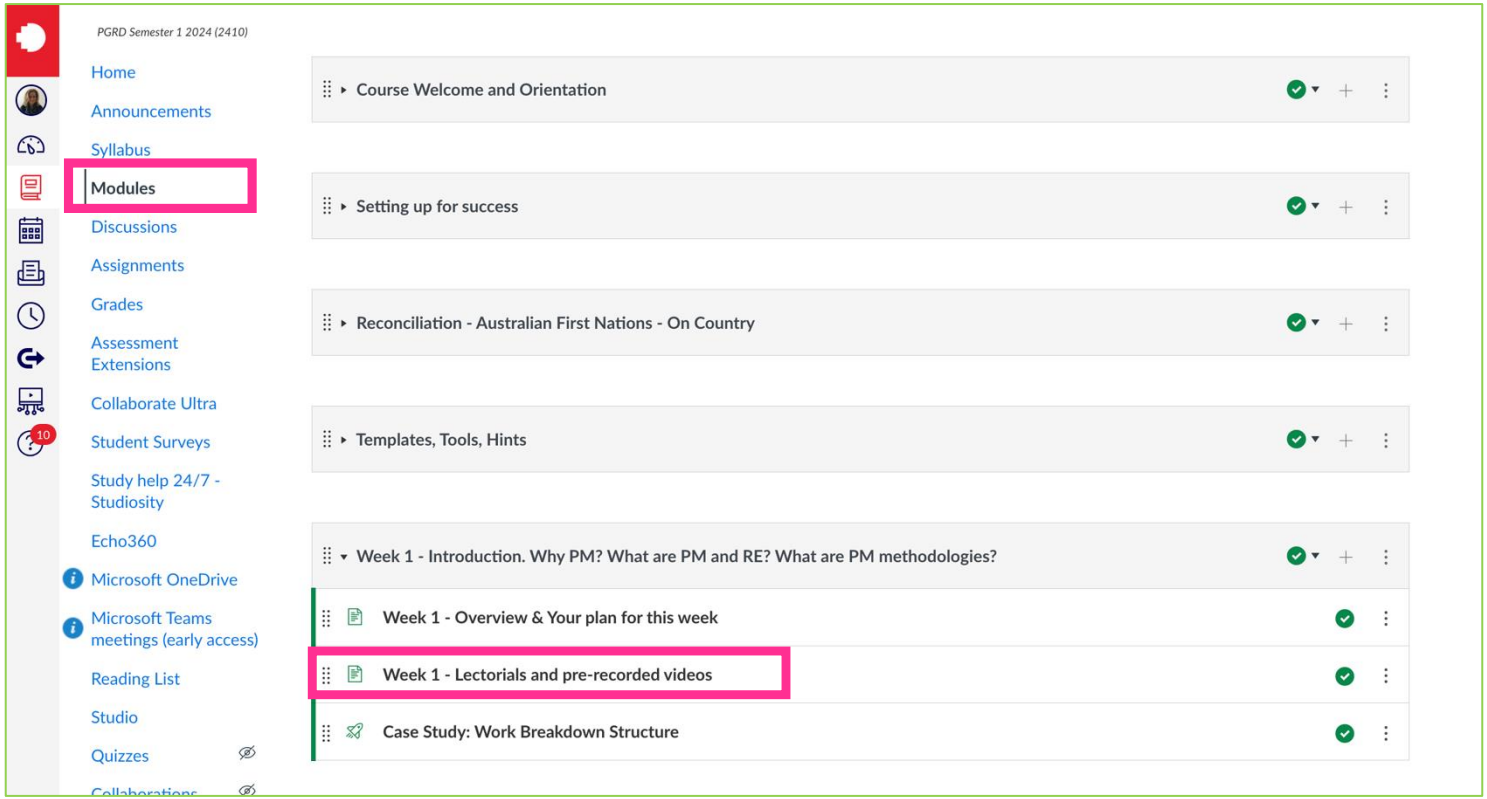
There is no Workshop on Week 1 but we shared with you a set of videos to watch

Questions?

Let’s go to www.menti.com ☺

Discussion on pre-recorded materials

Pre-lectorial materials



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Studio
Quizzes
Collaborations
People
Files
Outcomes
Rubrics
Pages
Item Banks
New Analytics
Library and Study

Workshops

Workshops start from Week 2. There are no workshop sessions this week :)

For this week we prepared for you instead a set of videos, which is compulsory to watch.

Watch

Before attending your Lectorial session, please watch the prerecorded videos on the following topics:

General introduction to PM:

1. Projects and project management (approx. 6 minutes)
2. Why projects fail? (approx. 8 minutes)
3. Requirements and stakeholders - core definitions (approx. 5 minutes)
4. More on requirements and their types (approx. 8 minutes)
5. Goals of project management - scope, time, cost, quality (approx. 2 minutes)

Vision and Scope document + Project Charter:

1. What is Vision and Scope document? (approx. 7 minutes)
2. Vision and Scope document - Business Requirements section (10 minutes)
3. Vision and Scope document - Vision section (approx. 6 minutes)
4. Vision and Scope document - Scope section (approx. 6 minutes)
5. Vision and Scope document - Business Context section (approx. 4 minutes)
6. What is Project Charter? (approx. 4 minutes)
7. Project Charter - Purpose and Objectives section (3 minutes)
8. Project Charter - Overview, Schedule and Resource Requirements sections (4 minutes)

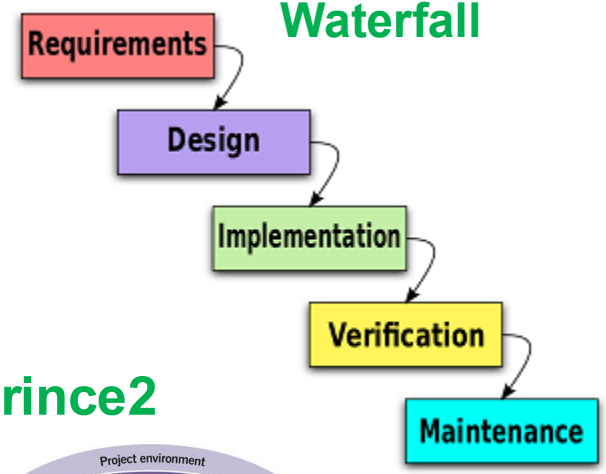
📚 Slides used for the videos on the lecture materials (*please watch the videos, they have more information that is possible to present on slides!*): SEPM_Week_1.pdf

Do you have any questions on these videos?

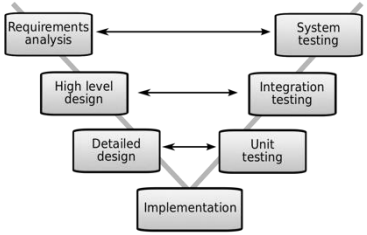
Project methodologies

Overview

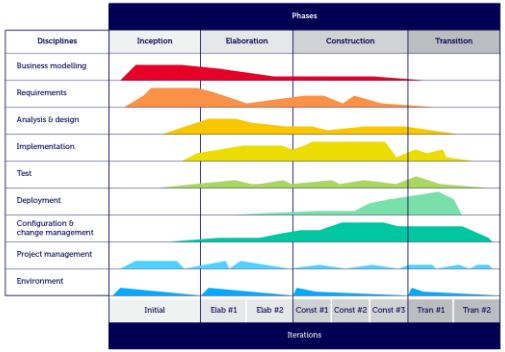
Waterfall



V-Modell



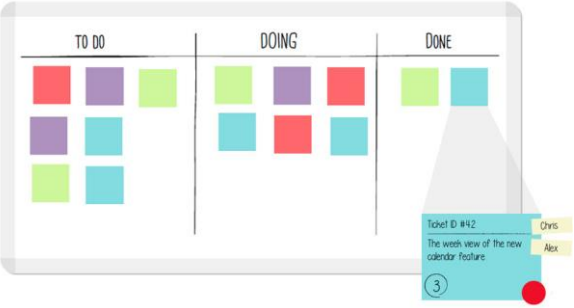
RUP



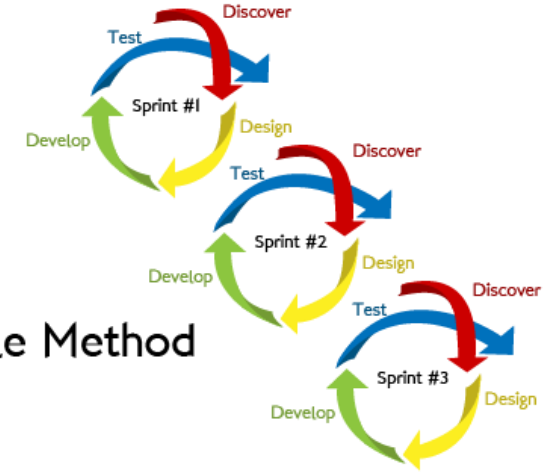
Prince2

LEAN

Kanban



Agile Method



Project Methodologies

Project Methodology:

- A process that documents a series of steps and procedures to bring about the successful completion of a project.
- A series of steps through which the project progresses

There are many Project Methodologies - variations of a them - common principles differently packaged:

- **Traditional – Waterfall**(-based)
- **RUP** (Rational Unified Process) introduced by Rational, a division of IBM
- **AUP** (Agile Unified Process) - simplified version of RUP
- **Agile**
- **Lean**
- **Prince2**
- …

They describe how the Software Development Life Cycles (SDLC) should work, i.e., how the phases of SDLC should be organized

Processes in SDLC

1. **Initiating Processes** - performed to define new project or phase of existing project by obtaining authorisation to start the project or phase
2. **Planning Processes** – performed to establish the total scope of the effort, define and refine objectives, and develop the course of action required to attain those objectives
3. **Executing Processes** – performed to complete the work defined in the project management plan to satisfy the project specifications
4. **Monitoring & Controlling Processes** – performed to track, review, and regulate the progress and performance of the project
5. **Closing Processes** – performed to finalise all activities across all Project Management Process Groups to formally close the project, phases, or contractual obligations

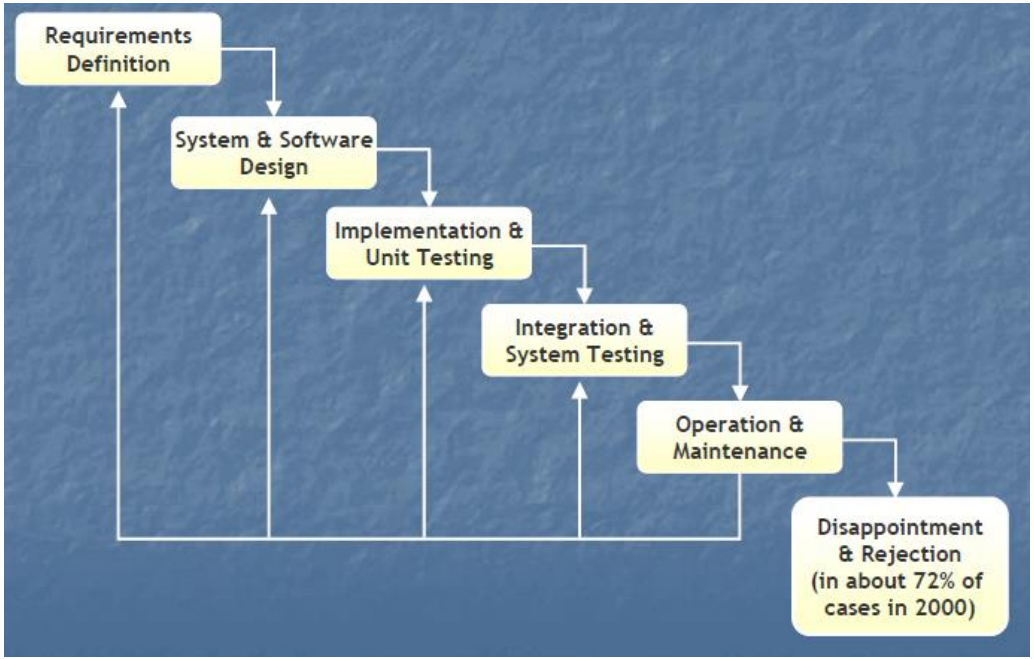
Waterfall

Waterfall = planning projects fully, then executing through phases

1956: the first mentioning of the idea by Herbert D. Benington at the Symposium on Advanced Programming Methods for Digital Computers

1970: the first formal description of the waterfall model in article by Winston W. Royce “Managing the Development of Large Software Systems”

1976: The earliest use of the term "waterfall" may have been by Bell and Thayer in their paper “Software requirements: Are they really a problem?”



Traditional Methodologies

- **Waterfall**
- **PMI's PMBOK** - applying universal standards to waterfall project management
- **PRINCE2** - waterfall project management methodology that includes principles, themes and processes (more strict)

- Linear sequential model
- Requirements need to be locked
- Development only starts after planning
- Output of one step acts as input of next step
- Still widely used
- Separate and distinct phases of specification and development
- Separate and distinct use/application of resources

Not flexible to readily incorporate changes

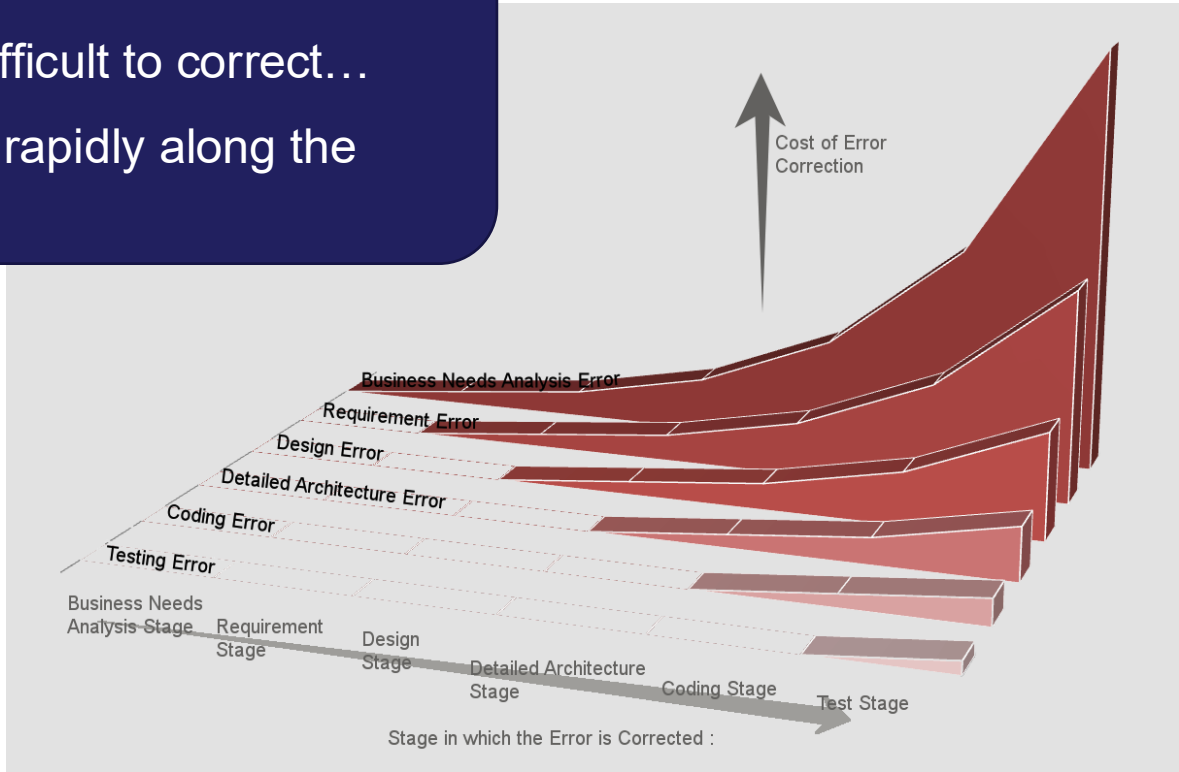
Inappropriate for version-wise release method

Error correction cost

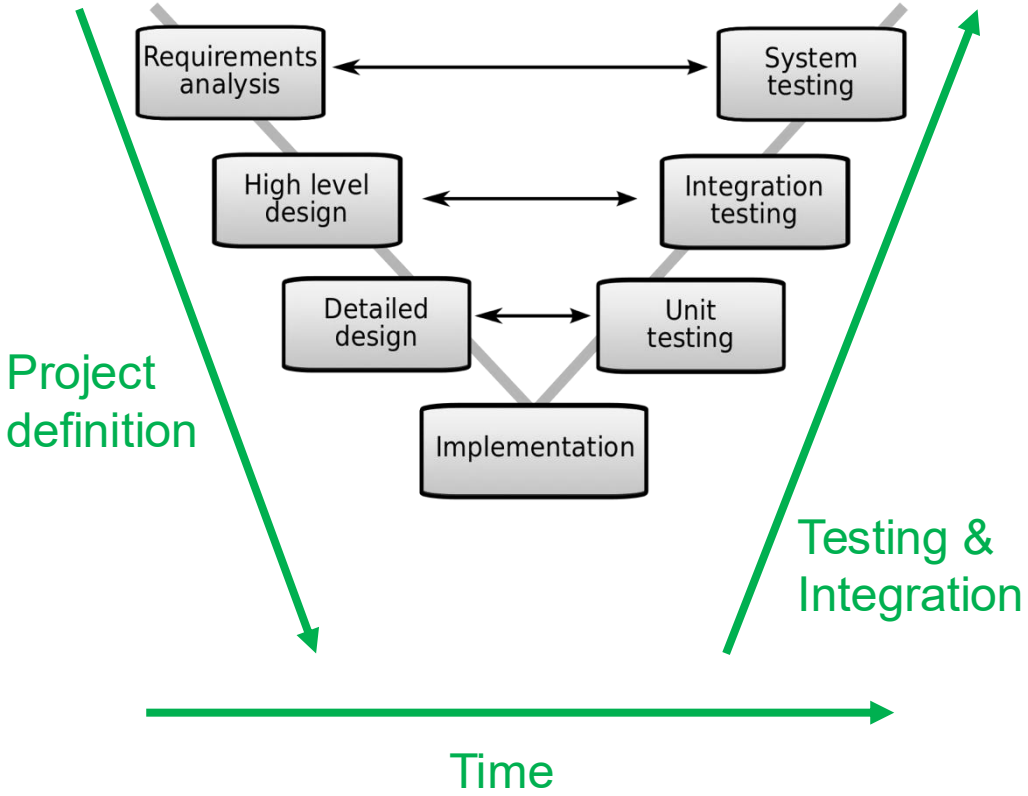
The failure / success cant be decided till too late.

Errors found in later phases difficult to correct…

Cost of error correction grows rapidly along the project life-cycle



V-Modell



A kind of improvement of Waterfall:

- ✓ High-level testing (verification) is done before coding
- ✓ Greater emphasis on testing
- ✓ Easy to understand and to use

- ✗ Still inflexible and linear
- ✗ Good for managers, but what about developers?

Is Waterfall really so useless?
Let’s discuss…

Agile

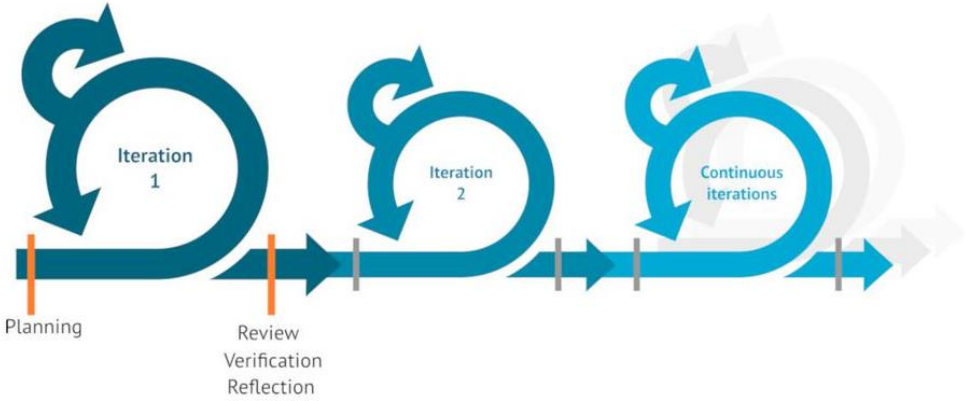
A framework for delivering products quickly and efficiently.

It encompasses set of exciting new practices and techniques that make product development more cyclical (or ***iterative***).

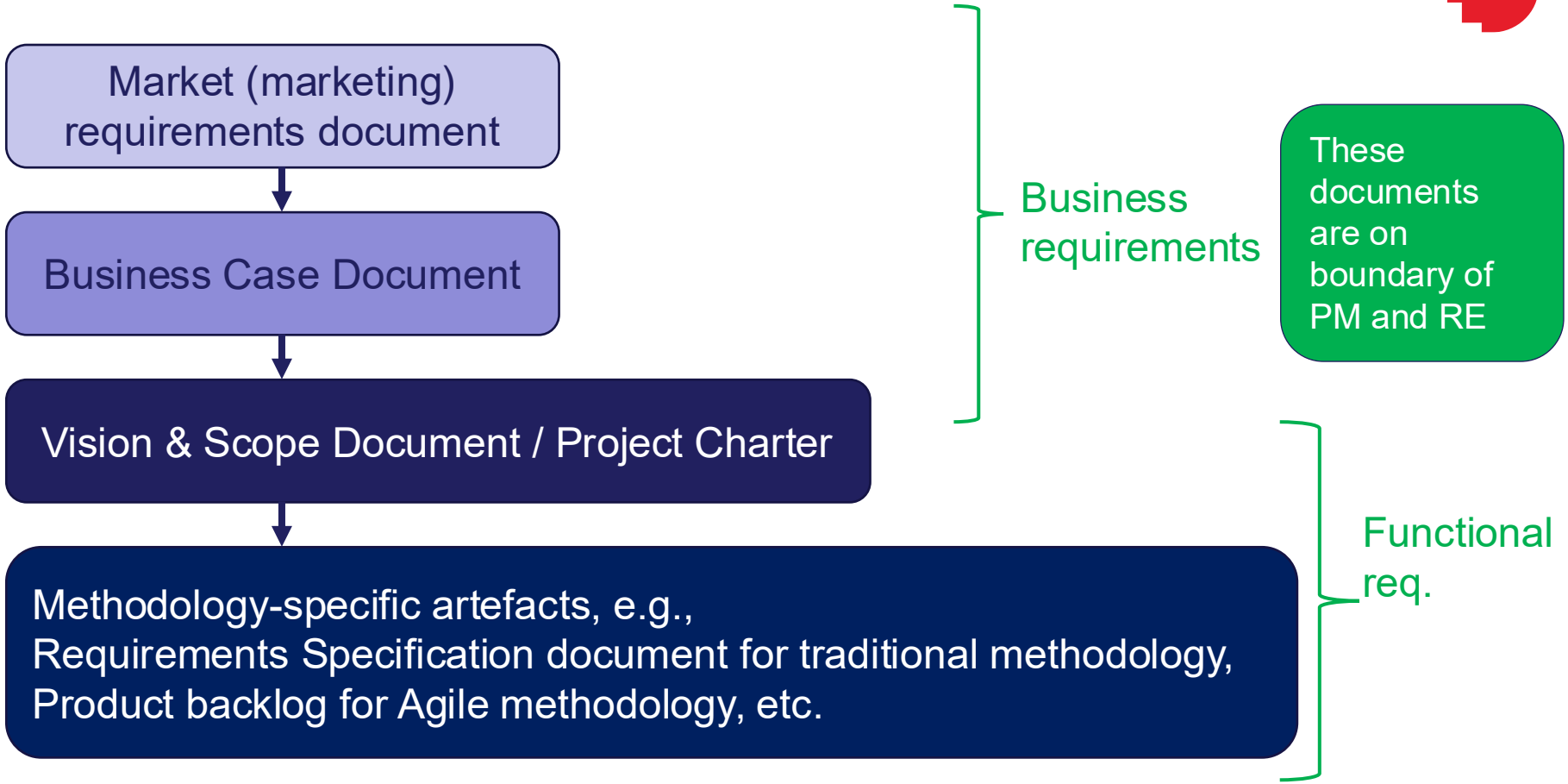
It relies on **lean governance (management)** as opposed to more traditional techniques that rely on **heavyweight governance**.

Agile is all about empowering the team and getting closer to what the customer wants.

As opposed to the standard requirements-gathering, plan-driven project development, the customer is continually involved in Agile.



Core artefacts we have to create at the beginning of a project



Vision & Scope Document / Project Charter

Both documents serve the same purpose:

- Collect all business requirements into a single deliverable that sets the stage for subsequent development work
- Define the business case for the project in nontechnical language (overlaps with Business Case document!).

- **Owner of this document:** project's executive sponsor, funding authority, …
- **Creator:** Business Analyst, but might be also Project Manager or Product Owner

Goals: Specify and justify the project

Goal/aim of these documents are the same, difference is in the format and focus on particular topics

→ We never use both documents in the same project

→ We have to decide at the very beginning, what format we follow

V&S Document vs. Project Charter

Vision & Scope document:

1. **Business Requirements**
   1) Background
   2) Business Opportunity
   3) Business Objectives and Success Criteria
   4) Business Assumptions and Dependencies
   5) Business Risks
2. **Vision of the Solution**
   1) Vision Statement
   2) Major Features
3. **Scope and Limitations**
   1) Scope of Initial and Subsequent Releases
   2) Limitations and Exclusions
4. **Business Context**
   1) Stakeholder Profiles
   2) Project Priorities
   3) Operating Environment

Project Charter (PMBOK)

1. Purpose *
2. Objectives *
3. Overview *
4. Schedules *
5. Resource requirements
6. Personnel and stakeholders *
7. Evaluation methods
8. Risk management *

What if we miss something in Project Charter / Vision & Scope document?

WBS: Traditional vs. Agile

WBS in Traditional / Waterfall projects

- Very detailed
- Who is doing what and when

Can we do the same in Agile?

Agile: You start with a high level plan and end up with a detailed plan…
So can we have WBS in Agile?

WBS in Agile:

- Agile WBS == your release plan has to be detailed
- Dependencies among features/ epics
- Traceability among features, user stories, and the sprint tasks
- (!) No allocation in advance to particular team members

WBS: Traditional vs. Agile

Traditional: Detailed, phase-oriented

TASK ID	TASK TITLE	PREDECESSOR	RESOURCE	START DATE	DUE DATE	DURATION

Agile: High-level, feature-oriented, more similar to a release planning

TASK ID	TASK TITLE	PREDECESSOR	DUE DATE

WBS: Traditional view – Exercise

A draft WBS was created, see below (the milestones are highlighted with (*)).
In this draft, some elements are missing or incorrect in this WBS.
Briefly explain what should be corrected and how.

ID	Task Name	Estimated Duration	Resource Name
1	System Design	2 Weeks	Jess, Alex
1.1	Design of the system architecture	2 Weeks	Alex
1.2	UI design	2 Weeks	Jess
1.3	Usability testing	1 Weeks	Anna
2	Product Backlog development	2 Weeks	Jess
3	Implementation (*)	8 Weeks	Alex, John, Max
4	Integration and System Testing	1 Weeks	Anna, John, Max
5	Documentation: User Guide (*)	2 Weeks	Jess
5.1	First draft of the User Guide	2 Weeks	Jess
5.2	Final version of the User Guide	1 Week	Jess

Project duration is expected to be 3 months, but the start date isn't specified yet.

The team includes:
Jess (requirements engineer),
Alex (system architect, developer),
John (developer, QA expert),
Anna (UI designer, QA expert),
Max (QA expert).

Missing/incorrect points?

Similar type of task will be in your A1.1 – SE challenge

WBS: Traditional view – Exercise

ID	Task Name	Estimated Duration	Resource Name
1	System Design	2 Weeks	Jess, Alex
1.1	Design of the system architecture	2 Weeks	Alex
1.2	UI design	2 Weeks	Jess
1.3	Usability testing	1 Weeks	Anna
2	Product Backlog development	2 Weeks	Jess
3	Implementation (*)	8 Weeks	Alex, John, Max
4	Integration and System Testing	1 Weeks	Anna, John, Max
5	Documentation: User Guide (*)	2 Weeks	Jess
5.1	First draft of the User Guide	2 Weeks	Jess
5.2	Final version of the User Guide	1 Week	Jess

Missing/incorrect points:

1. Missing predecessors
2. Missing RE phase
3. Usability testing cannot be done before implementation
4. Usability testing cannot be a part of system design
5. Product backlog is an Agile/ Scrum artefact, but the whole WBS is traditional
6. Duration of Task 5 should be 3 weeks = 2 weeks + 1 week (5.2 can be done only when 5.1 is completed), if we really keep durations of 5.1 and 5.2 as they are specified – better to make them shorter, 2-3 days each
7. Too short time for testing
8. 3 months = approx. 12 weeks, where the overall sum of tasks' durations is much longer
9. Task 1.2 should be done by Anna instead of Jess, Task 1 will be then done by Anna and Alex

WBS: Traditional view – Exercise (continues)

ID	Task Name	Estimated Duration	Resource Name
1	System Design	2 Weeks	Jess, Alex
1.1	Design of the system architecture	2 Weeks	Alex
1.2	UI design	2 Weeks	Jess
1.3	Usability testing	1 Weeks	Anna
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5.1	First draft of the User Guide	2 Weeks	Jess
5.2	Final version of the User Guide	1 Week	Jess

Corrected WBS:

ID	Task Name	Pred.	Estimated Duration	Resource Name
1	Requirements Specification (*)	-	2 Weeks	Jess
2	System Design	1	2 Weeks	Anna, Alex
2.1	Design of the system architecture	1	2 Weeks	Alex
2.2	UI design	1	2 Weeks	Anna
3	Implementation (*)	2	4 Weeks	Alex, John, Max
4	Integration and System Testing, Usability testing	3	3 Weeks	Anna, John, Max
5	Documentation: User Guide (*)	4	1 Week	Jess
5.1	First draft of the User Guide	4	2 days	Jess
5.2	Final version of the User Guide	5.1	3 days	Jess

WBS: Agile view – Example (fragment)

Analyse the difference between Agile and Traditional WBS on this example

ID	Name	Predecessor	Due date
1 (*)	Web application: Version 1		1 April
1.1	Order lunch from cafeteria - payroll		
1.2	Create and view menus		
1.3	Web Login / Logout		
1.4	Web Registration		
2 (*)	Web application: Version 2	1	1 June
2.1	Order lunch from cafeteria - credit and debit card payments	1.1	
2.2	Order lunch from restaurants to campus only	1.1	
2.3	Add, edit, and delete menus	1.2	
2.4	Ingredients list	1	
3 (*)	Android and iOS applications: Version 1	1	1 June
3.1	Order lunch from cafeteria - payroll	1	
3.2	Order lunch from cafeteria - credit and debit card payments	1	
3.3	Order lunch from restaurants to campus only	1	
3.4	App Login / Logout	1	
3.5	App Registration	1	
4 (*)	Windows phone applications	3	1 August
4.1	Order lunch from cafeteria - payroll	3	
4.2	Order lunch from cafeteria - credit and debit card payments	3	

PGRD Semester 2 2024 (2450)

Account

Dashboard

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Week 1: Introduction. Why PM? Project Charter vs. Vision & Scope Document

Week 1 - Overview & Your plan for this week

Week 1 - Lectorials and pre-recorded videos

Case Study: Work Breakdown Structure

Case Study: Work Breakdown Structure

Due No due date **Points** 5 **Questions** 1 **Time Limit** None
Allowed Attempts Unlimited

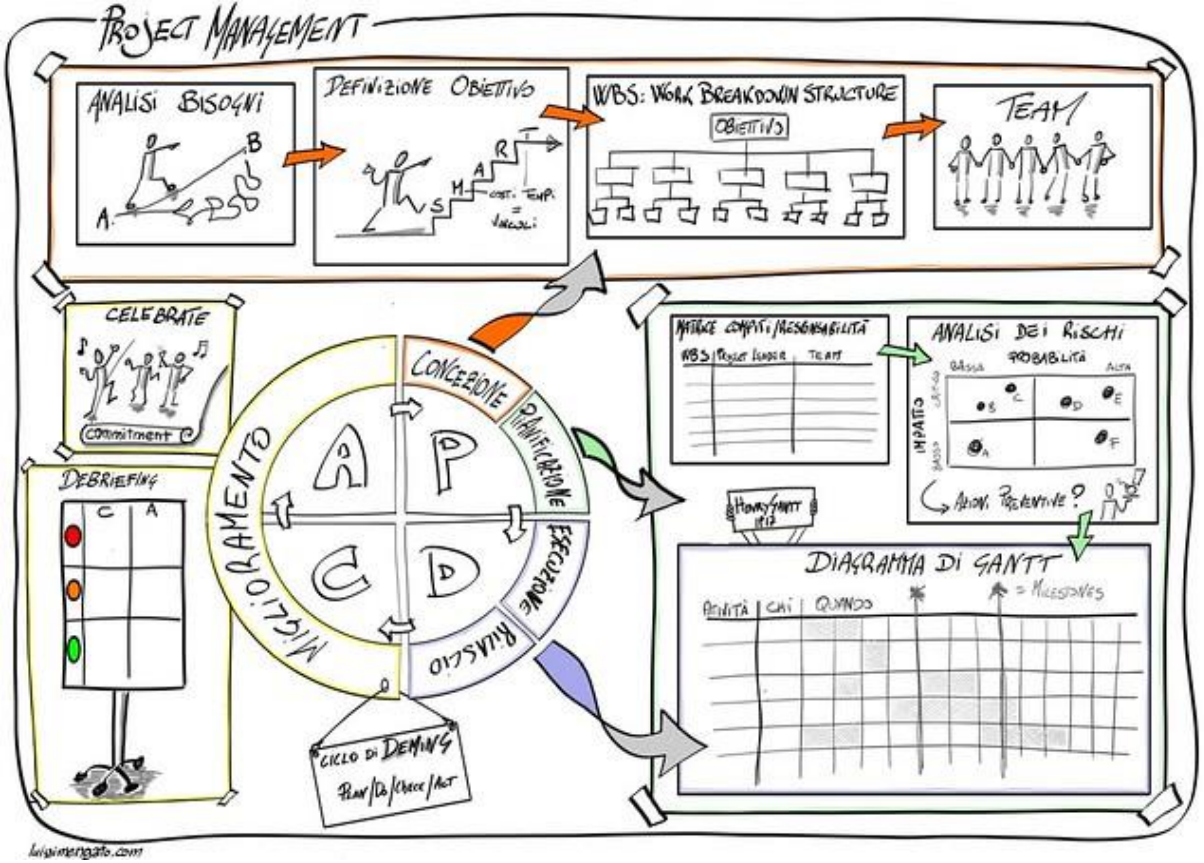
Instructions

This is ungraded, unlimited attempt quiz for you to self-check your knowledge of WBS and traditional project planning.

A detailed analysis of the provided WBS and required changes are discussed in Week 2 Lectorial.

Take the Quiz

Questions?


PROJECT MANAGEMENT
ANALISI BISOGNI
DEFINIZIONE OBIETTIVO
WBS: WORK BREAKDOWN STRUCTURE
OBIETTIVO
TEAM
CELEBRATE
COMMITMENT
DEBRIEFING
MIGLIORAMENTO
CONCEZIONE
PIANIFICAZIONE
ESECUZIONE
RILASCIO
CICLO DI DEMING
PLAN/DO/CHECK/ACT
MATRICE COMPITI/RESPONSABILITÀ
ANALISI DEI RISCHI
PROBABILITÀ
IMPATTO
AZIONI PREVENTIVE?
HOWTOGANTT
DIAGRAMMA DI GANTT
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